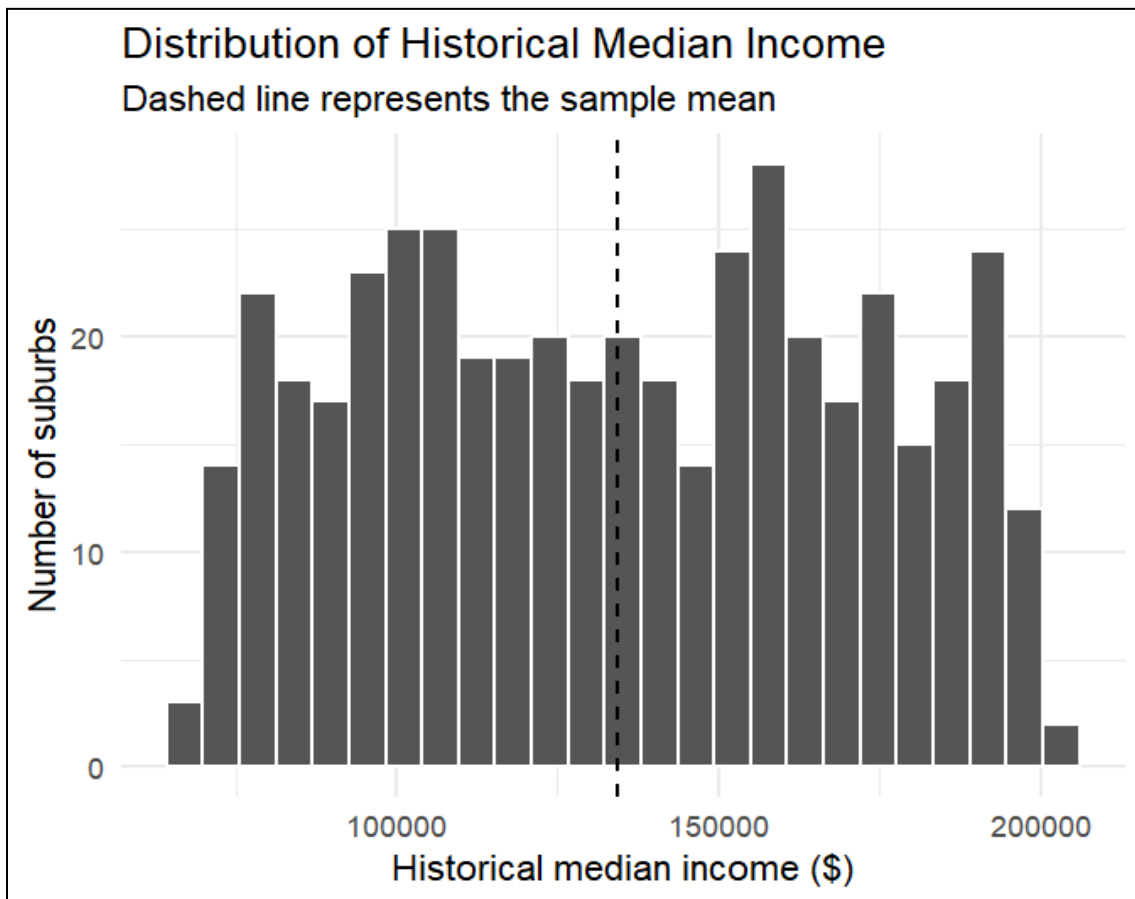


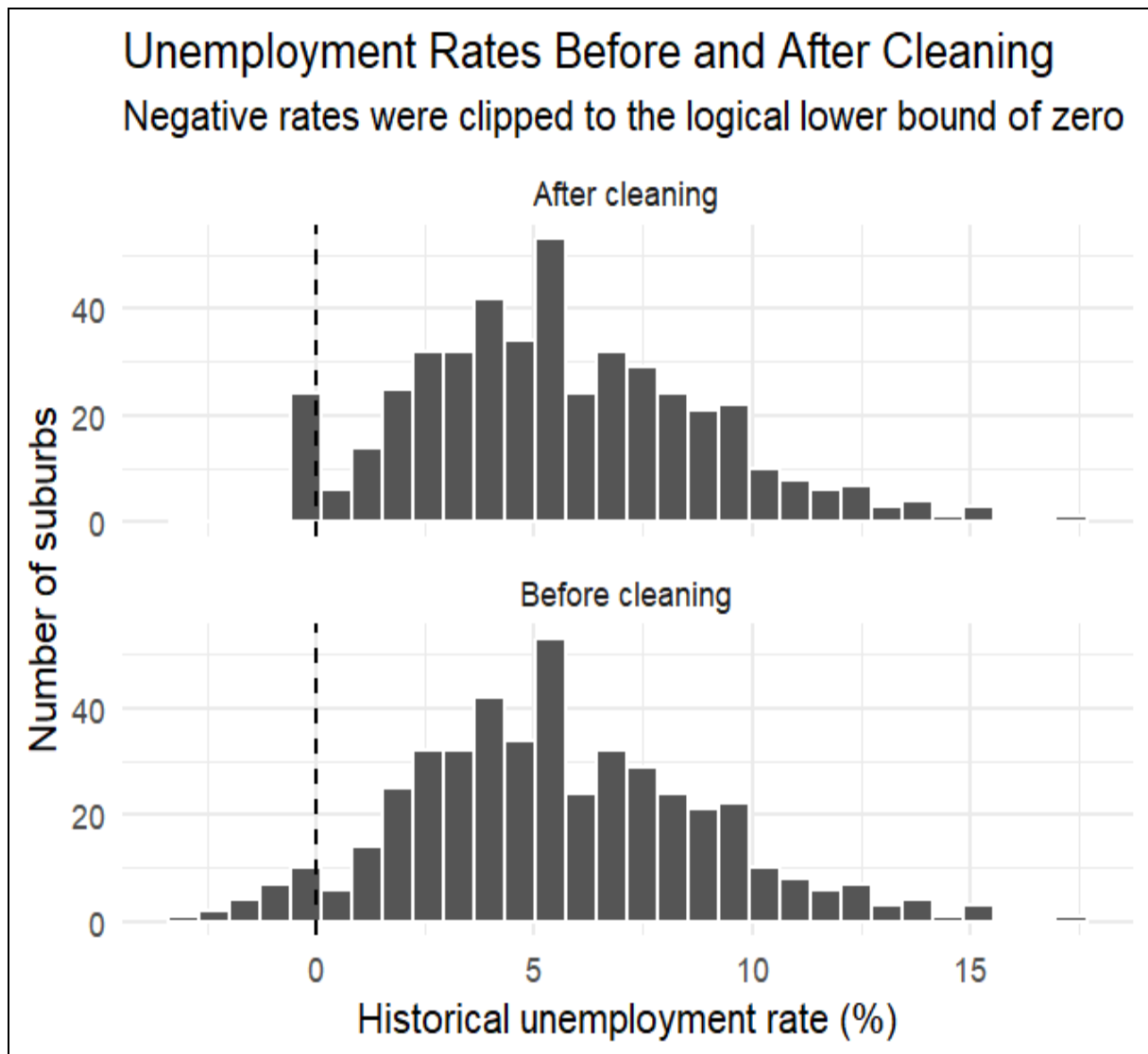
Victorian Apartment Investment

1. Data Summary

The three given datasets contain apartment prices and (historical or projected) information for 458 unique Victorian suburbs. "**Suburb_name**" is the common join key, with no duplicate or unmatched records. **NORLANE**'s price was recorded as "309334x". The extra character was removed, and the remaining value was converted to numeric. **POINT LONSDALE** had one missing historical median income value. The income distribution was balanced, and the mean (\$134,322) was close to the median (\$134,817). So, the missing value was replaced with the mean.



Additionally, "22" historical and projected unemployment rates were negative. Unemployment rate cannot fall below zero. So, the negative values were set to zero while the positive values for unemployment were retained as in the raw data.



MAFFRA's price of **\$4.20 million** exceeded the three-IQR upper threshold of \$871,211 and was over six times the next highest price. It was excluded from modelling to prevent disproportionate influence (*outlier*), leaving "457" suburbs.



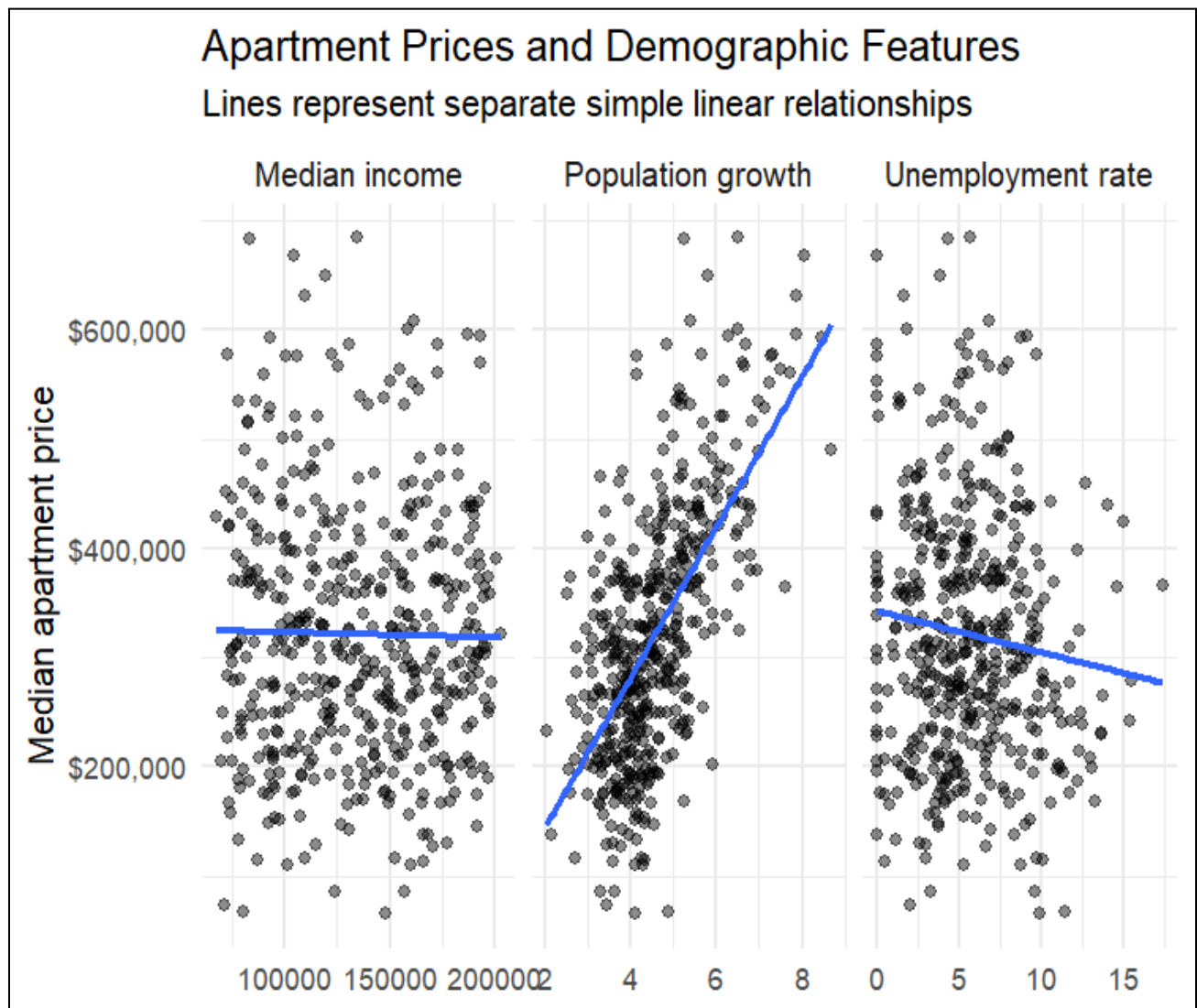
2. Model Estimation

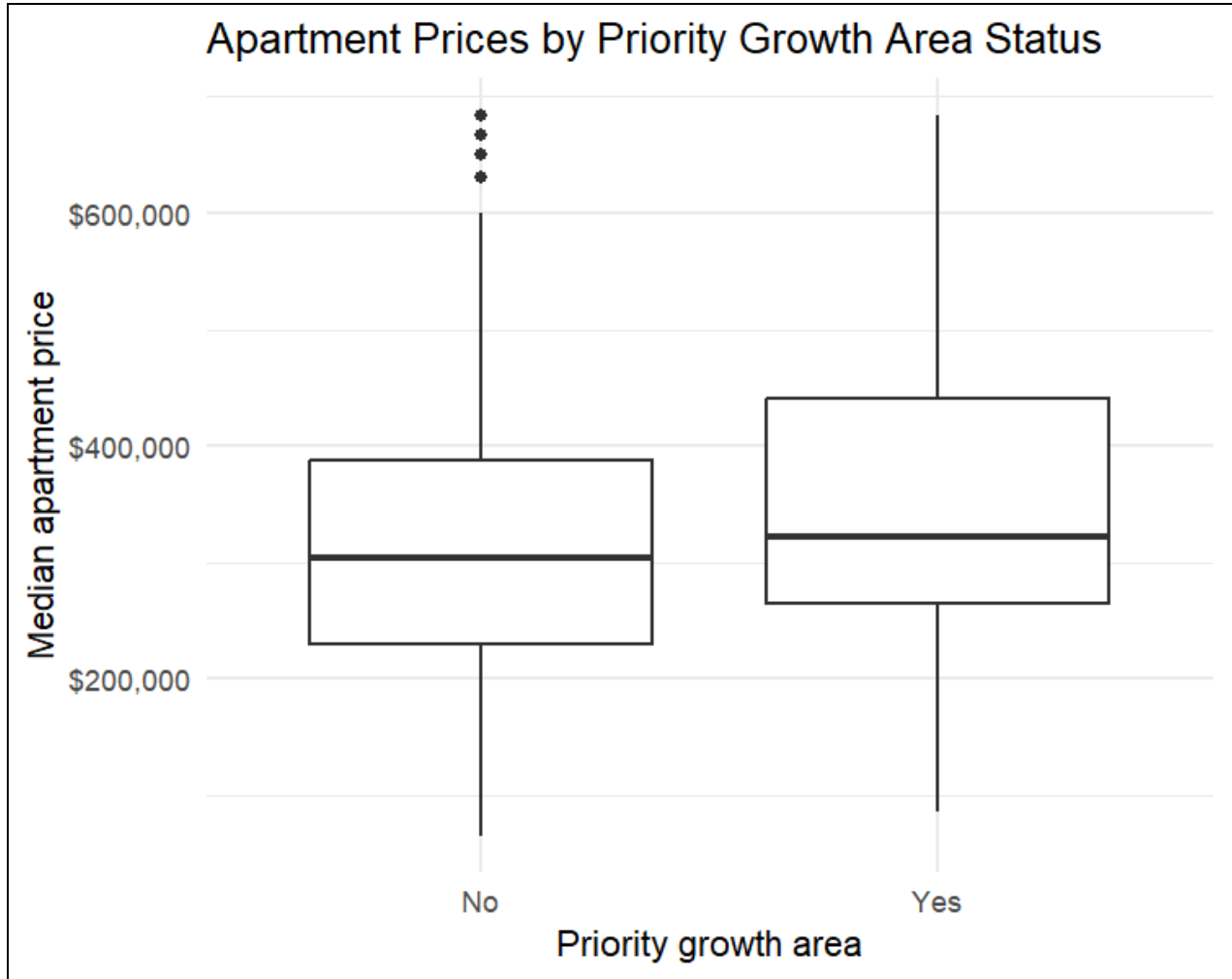
The initial linear model equation :

$$Price_i = \beta_0 + \beta_1 PopulationGrowth_i + \beta_2 UnemploymentRate_i + \beta_3 PriorityGrowthArea_i + u_i$$

The Graphs below suggest a very strong positive relationship between **population growth and price**, a negative relationship with unemployment, and higher prices in priority-growth areas.

Median income showed little to no relationship with price.



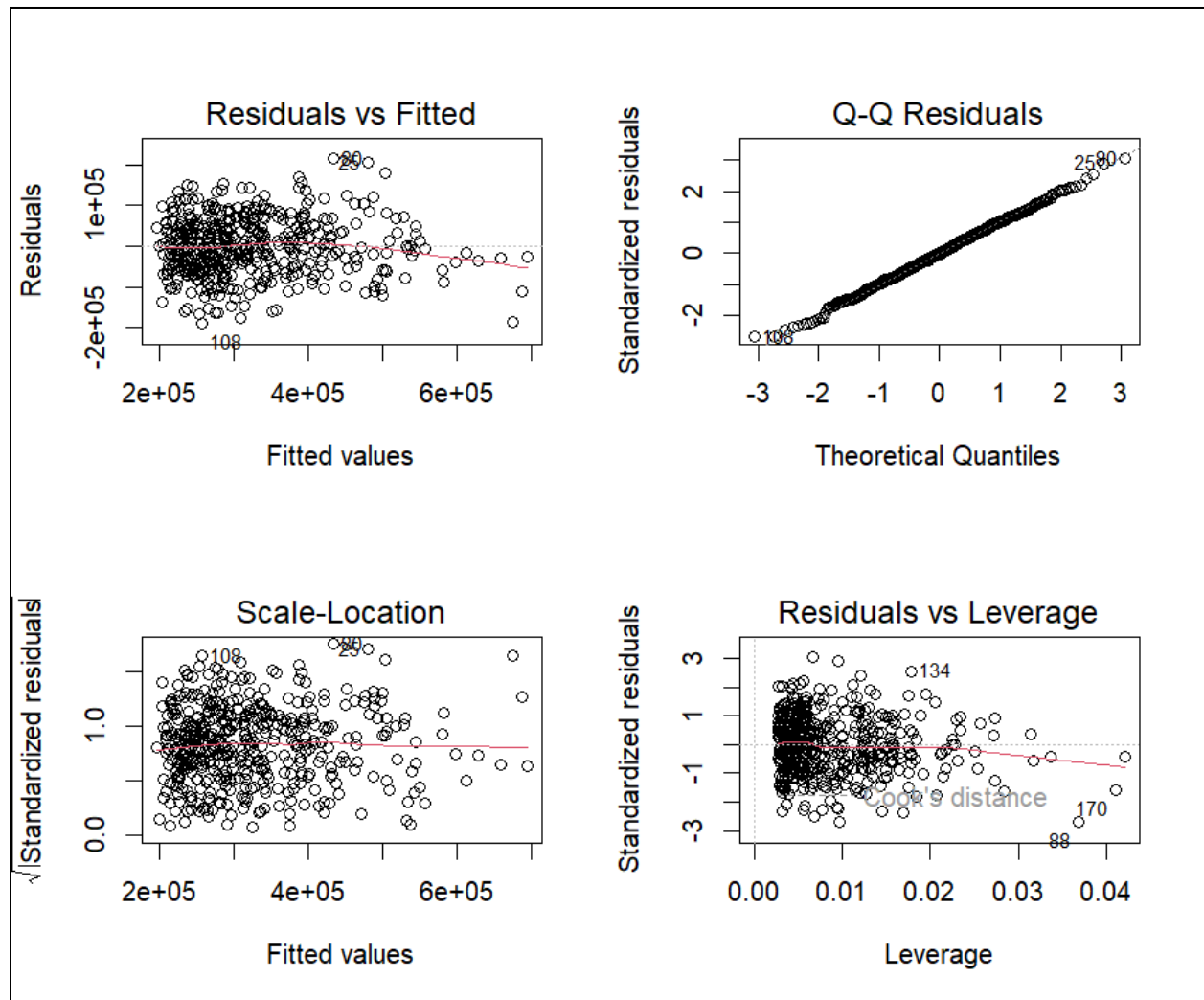


The full model explained **64.4%** of price variation, but the median income was statistically insignificant ($p=0.133$). Its 95% confidence interval included zero, and a nested-model comparison showed that adding it did not significantly improve fit ($p=0.1334$). Median income provided insufficient additional or negligible explanatory value and was removed from the preferred model.

So, the reduced model was selected:

$$\widehat{Price}_i = -69,435 + 95,159PopulationGrowth_i - 12,163UnemploymentRate_i + 125,558PriorityGrowthArea_i$$

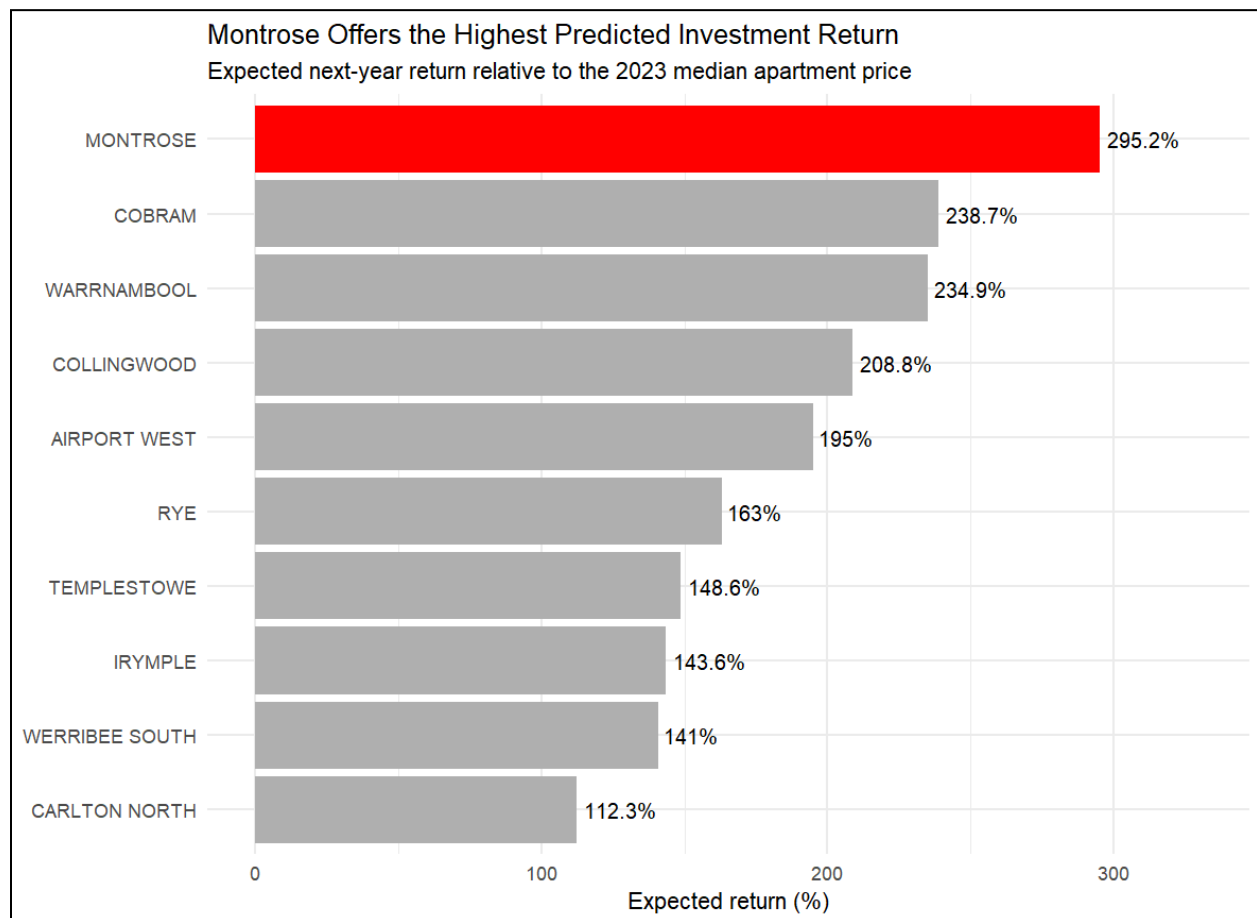
All remaining predictors were significant ($p < 0.001$), and the model explained **64.2%** of price variation. Diagnostic plots showed approximately normal residuals and no severe leverage problem, although "*mild heteroskedasticity*" may remain. Excluding SEAFORD, the most influential suburb, also produced similar coefficients and model fit. All of this supports robustness. This follows our course's ceteris paribus interpretation of multiple regression.



3. Model Interpretation

If we hold other factors fixed, then a one-percentage-point increase in population growth was associated with a **\$95,159** increase in apartment price. A one-percentage-point increase in unemployment was associated with a **\$12,163** decrease, while priority-growth-area suburbs were predicted to be worth **\$125,558 more** than otherwise comparable suburbs.

“Best” was defined as the highest predicted percentage return relative to the current median price. Applying the model to projected demographic data identified **MONTROSE** as the strongest opportunity. Its median price was forecast to rise from \$67,157 to approximately \$265,413, an expected return of 295.2%. MONTROSE also ranked first using the lower 95% prediction bound, which still implied an 87.2% return. Its current price was not a lower outlier, so the result was not driven by an obvious data error.



The predictor variables were not strongly correlated with each other, so multicollinearity was not an issue. However, relationships between demographic factors and apartment prices aren't perfectly linear, and suburbs may not represent a random sample. The model also does not include factors like interest rates, housing supply, etc. These omitted factors may influence demographic conditions and apartment prices. The diagnostic plots also suggested mild heteroskedasticity, and the unemployment corrections may cause measurement error. Regardless, **MONTROSE** remains the strongest investment recommendation.